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Measuring Liquidity

Learning Objectives:

- The components of trading costs
- How implicit trading costs can be measured from quote data
- What to do when only transaction data are available
- How to take the time dimension in order execution into account

2.1 Introduction

If the structure of a securities market is compared to a car design, measuring market liquidity can be likened to assessing the car's driving performance. Several different aspects of performance need to be considered (for cars, fuel efficiency, speed, safety, etc.), and liquidity has several dimensions as well: trading costs, depth available to customers placing large orders, speed of execution, protection against execution risk, and so on. Liquidity measurement is of paramount importance to practitioners, since illiquidity affects portfolio returns, as Chapter 10 explains. Investors and intermediaries accordingly want to design trading strategies that minimize the effects of illiquidity on their investment performance. Liquidity measurement is also important for researchers and regulators who need to understand the relationship between market structure and performance, just as engineers are keen to understand the relationships between various aspects of a car's design and its road performance.

Illiquidity is often gauged by the cost of trading, which has both an explicit and an implicit component. Explicit costs include broker commissions, transaction taxes, platforms' trading fees, and clearing and settlement fees. As they are charged explicitly to final investors, these are easy to measure. Implicit trading costs are those that arise from the illiquidity of the market; they are measured by the gap between execution price and some benchmark used to proxy for the price that would be obtained in a perfectly liquid market. Commonly, where the data are available, this benchmark is the midquote (the average of the best bid and ask prices) at the time the order is placed or executed.

In practice, professional traders pay close attention to trading costs, and some brokers periodically report summary measures of both components based on the orders executed. For instance, Virtu Financial releases these measures by country for each quarter, as shown in Table 2.1.

The last column in Table 2.1 reports the total trading costs for orders executed by Virtu's brokerage arm in Europe (excluding the United Kingdom) for each quarter (expressed in basis points, i.e., as hundredths of a percent of the value traded).¹ The first three columns break this total down into its components. Implicit trading costs consist of delay costs (second column) and impact costs

Table 2.1 Trading costs (in bps) in European Equity Markets (Ex. U.K.)

Quarter	Delay Costs	Impact Costs	Comm. Costs	Total
2016				
Q1	26	15	8	49
Q2	17	16	8	40
Q3	20	13	8	41
Q4	19	13	7	40
2017				
Q1	17	11	7	35
Q2	12	12	7	31
Q3	15	12	7	34
Q4	16	13	7	36
2018				
Q1	16	12	5	34
Q2	20	11	5	36
Q3	24	12	5	41
Q4	27	15	5	46
2019				
Q1	18	14	5	37
Q2	16	14	5	35
Q3	19	13	5	37
Q4	18	13	5	36
2020				
Q1	24	22	5	51
Q2	22	28	4	54
Q3	24	15	4	43

Source: Virtu Global Cost Review, 2020, available at <https://www.virtu.com/thinking/thought-leadership/>.

¹ One basis point (bp) is 0.01%. Thus, a total trading cost of 50 bps means that for a trade of \$100, the investor pays a total trading cost of 50 cents. This may seem a small amount but it is not if one considers that the trading volume in securities markets is often large. For instance, in August 2020 the total value traded on the LSE's U.K. order book was £66 billion (source: LSE website). In this case a 50 bps trading cost implies a total illiquidity cost of about £330 million.

(third column). For a buy order, the delay cost is the difference between the midquotes at order execution and submission times; the impact cost is the difference between execution price and midquote at execution time. Explicit trading costs (commissions and fees) are shown in the fourth column as “Comm Costs.”

Several interesting facts emerge. First, explicit trading costs are much smaller than implicit trading costs. Indeed, commission costs have declined steadily, continuing a long-term pattern: relative to the mid-2000s, they have dropped by two-thirds. Second, within the implicit costs, delay costs are relatively high, because on average prices move adversely while execution is delayed. Finally, implicit trading costs were largely stable, but rose during periods of market volatility, such as the first and second quarter of 2020, when markets were roiled by the coronavirus pandemic.

Just as someone choosing a car seeks comparative indicators of its performance, traders are interested in measures of implicit trading costs. For instance, a broker who needs to buy five thousand shares of Microsoft will compare its price on Nasdaq (where the stock is listed) to its price on several other platforms where it is traded (Cboe BZX, NYSE, etc.). He must then decide how many shares to buy in each market; whether to split the order over time; and whether to use market orders, limit orders, or a combination of the two.

In this situation, information on implicit trading costs is useful: for instance, the broker may want to trade more on the platform with lower implicit trading costs. Estimating this cost requires detailed data on quotes (e.g., the full LOB in each trading venue) and on past orders (such as the time of their submission, to measure delay cost). When such data are available, traders can assess market liquidity via simple measures of implicit trading costs, such as the quoted bid-ask spread, the effective bid-ask spread, and the realized bid-ask spread, which are presented in Section 2.2. In some cases, regulators oblige trading platforms to publish periodic data on some of these implicit cost measures.²

These measures of trading costs require data on bid and ask quotes, but these are sometimes simply unavailable. Even so, implicit trading costs can be measured using time series of recent transaction prices and possibly trading volume. In Section 2.3 we present these additional measures of implicit trading costs: the gap between transaction prices and the volume-weighted average price (VWAP); the estimated price impact of orders; measures of illiquidity based on non-trading; and Roll’s measure of illiquidity, which is based on the serial covariance of transaction price changes.

² Since 2000, the SEC has required platforms that trade stocks listed on Nasdaq or the NYSE to report monthly statistics (known as Dash-5 reports) on their execution quality. These statistics include the average effective bid-ask spread, the average realized spread, and the speed of execution for orders of various sizes and types.

All these measures, however, neglect the time dimension of implicit trading costs: if traders are willing to trickle their orders into the market gradually, they may get a better average price, because with time new counterparties may emerge and post new orders to replace those that have been exhausted—a dynamic dimension of liquidity known as market resiliency. But a gradual execution strategy exposes traders to the risk of incomplete execution. Section 2.4 discusses the notion of implementation shortfall, a measure of execution quality that considers not only the price impact of orders but also the opportunity cost of delayed or partial execution. Traders frequently use this measure to assess their total implicit trading costs on large orders (ex post, after the execution of these orders) and devise trading strategies for minimizing these costs.

2.2 Measures of the Spread

As explained in Section 1.2.1, the most direct way to measure implicit trading costs is to look at market quotes and do a “what if” experiment: what would it cost to make a round-trip transaction, that is, to buy and instantly resell a given amount of securities? Different spread measures give different answers.

2.2.1 The Quoted Spread

The obvious, intuitive measure of the cost of a *small* round-trip transaction is the difference between the best ask quote a and the best bid quote b , that is, the *quoted bid-ask spread* $S = a - b$. If this spread is normalized by the midprice $m = (a + b)/2$, one obtains the relative quoted spread:

$$s \equiv \frac{S}{m} = \frac{a - b}{m}. \quad (2.1)$$

The quoted spread is a good measure of trading costs for orders that are so small that they can be entirely filled at the best quotes, which in the United States are known as the Best Bid and Offer (BBO). The quoted spread for small trades is the most widely reported measure of illiquidity; this is what people mean when they refer to “the” bid-ask spread.

As we saw in Chapter 1, the spread on larger orders can be gauged in a similar way, computing a weighted average bid-ask spread from the quotes posted. For instance, for the buy and sell limit orders posted at a given point in time, suppose that the average execution price for a buy market order of size is $\bar{a}(q)$, and the average execution price for a sell market order of size is $\bar{b}(q)$. The average bid-ask

spread for an order of size q is thus $S(q) = \bar{a}(q) - \bar{b}(q)$ and the relative weighted average bid-ask spread is:

$$s(q) = \frac{\bar{a}(q) - \bar{b}(q)}{m}. \quad (2.2)$$

Clearly, when q is so small that the entire offer can be filled at the BBO, this reduces to the spread s in equation (2.1). As the quantity offered at each price is limited, $s(q)$ increases with the trade size q . The deeper the market, the milder the increase in the spread $s(q)$ associated with larger trade sizes q .

The drawback to this method is that it requires data on limit orders or quotes posted in the market at various points in time. Moreover, for trade sizes large enough to exhaust the liquidity offered at the BBO, the LOB at price points beyond the BBO is required. As this information is not always readily available, practitioners and researchers often measure implicit trading costs based on transaction prices alone, or on transaction prices together with the average of the BBO. In the rest of this section, we describe two such measures: the effective spread and the realized spread.

2.2.2 The Effective Spread

The quoted spread reflects the liquidity available at a given point in time for a hypothetical transaction. Instead, one could measure trading costs using the prices actually obtained by investors. This gauge is the *effective half-spread*, defined as the difference between the price at which a market order executes and the midquote on the market the instant before. Suppose that a market buy order for one thousand shares arrives and executes at an average price of 75.50, while the prevailing midquote is 75.45. The effective half-spread on to this order is thus $75.50 - 75.45 = 0.05$ or, as a percentage of the midquote, $0.05/75.45 = 0.066\%$. Formally, the absolute effective half-spread is defined as:

$$S_e \equiv d(p - m), \quad (2.3)$$

where d is the order direction indicator (1 for buyer-initiated and -1 for seller-initiated trades) and m is the midquote on the market prior to a transaction executed at price p . In relative terms, the effective spread is:

$$s_e \equiv d \cdot \frac{p - m}{m}. \quad (2.4)$$

The effective spread can be seen as a measure of a transaction's impact on the price, since it measures the deviation of the actual execution price from the

midprice prevailing just before the transaction.³ This impact (sometimes called “slippage”) is positive precisely because the liquidity of the market is limited. Thus, the effective spread averaged across a large number of transactions is a way to gauge market liquidity. The effective spread is likely to increase with the size of the transaction in that (as Chapter 1 explains) larger market orders execute at less favorable prices. So it is useful to sort transactions into size classes before averaging, and get a series of estimates of the effective spread, for various sizes.

As it is based on actual transactions, the effective spread captures any price improvement that market orders may receive when they execute against hidden orders or when they receive a price improvement in dealer markets, as explained in Chapter 1. However, being based on past prices, the effective spread is a retrospective measure of liquidity, in contrast with the quoted spread that relies on prices at which traders can actually trade at a given point in time.

As shown by equation (2.3), to calculate the effective spread one needs transaction prices, the midprice prior to each transaction, and an order direction indicator (buy or sell initiation).

In practice, performing this calculation can be difficult for at least two reasons. First, in theory the effective spread for an order that is split over time should be measured by comparing the average price over the entire order with the market midquote at the time of the first part of the transaction. For an investor assessing the quality of execution of a completed trade, it is a simple matter; one can readily determine the average price—even if the order is executed at multiple price points and/or involves some use of limit orders—and compare it to the initial market midprice. And analyzing such data can help evaluate the effectiveness of trading strategies: whether splitting orders and trickling them into the market improves the average price, and how slowly large orders should be worked. For an econometrician or a regulator analyzing liquidity, however, the splitting of transactions can make it nearly impossible to reconstruct total orders from transaction records. Thus the analysis of market liquidity by such a non-participant may be restricted to the effective spread for smaller transactions placed by an identifiable subset of traders, such as market order placers.

Second, some data sets do not tell whether transactions stem from buy or sell market orders. When this information is not available, one must devise a way to “sign” transactions, that is, to decide for each trade whether $d = 1$ or $d = -1$. One of the most commonly used and accurate methods is the algorithm proposed by Lee and Ready (1991), which is described in Box 2.1.

³ Ideally, one would like to measure trading costs by the difference between the price at which liquidity demanders trade and the fair value of the asset when they trade—the “true” effective bid-ask spread. However, this true effective bid-ask spread cannot be measured directly because the fair value is not observed. Hagströmer (2021) shows that the effective bid-ask spread measured relative to the midquote overestimates the true effective bid-ask spread because liquidity demanders are more likely to submit buy (sell) market orders when the fair value is above (below) the midquote.

Box 2.1 Classification of Trades

The Lee-Ready algorithm is commonly used to classify transactions as buys or sells when this information is not available in the data set. The algorithm works as follows. A transaction is classified as “buyer-initiated” if its price is closer to the prevailing ask quote than to the bid, and as “seller-initiated” if the converse. Any transaction priced exactly at the midquote is classified as a buy if the price is higher than the previous transaction price (that is, on an “uptick”), a sell if lower. The Lee-Ready algorithm and any of its subsequent refinements cannot classify trade initiation with perfect accuracy and so may bias measured transaction costs. Using data sets in which trade directions are observed, Chakrabarty, Pascual, and Shkilko (2015) and Panayides, Shohfi, and Smith (2019) find that the Lee-Ready algorithm is accurate in around 85% and 79% of cases, respectively (that is, about 20% of the trades are misclassified). Jurkatis (2022) proposes a new classification algorithm that outperforms the Lee and Ready algorithm in fast markets (see also Holden and Jacobsen 2014).

2.2.3 The Realized Spread

The quoted spread and the effective spread implicitly adopt the viewpoint of liquidity demanders. That is, they gauge the extra cost sustained by a trader submitting a market order relative to an ideal environment in which trades are made at the midprice. It is tempting to conclude that this extra cost is a gain for liquidity suppliers, the counterparties to all trades by liquidity demanders, but this is not actually the case, because buy and sell orders may exert lasting pressure on prices, to the detriment of liquidity suppliers (e.g., dealers) who absorbed such orders into their inventories.

To illustrate, suppose that a dealer buys 75 shares at \$326 when the best bid and ask prices are \$326 and \$327, respectively. If the dealer unwinds his position immediately at the ask price, he earns a profit of \$1 on each share. But if the best bid and ask prices decline to, say, \$325.5 and \$326.5, respectively, he earns a zero average profit (assuming that he is equally likely to unwind his inventory at the best ask or the best bid price). This example shows that quoted or effective bid-ask spreads are likely to overestimate liquidity providers’ gains (and therefore liquidity demanders’ trading costs) if after a trade prices move in the direction of the trade.⁴

⁴ One reason for this is that some market orders may be placed by investors with advance information (see Chapters 3 and 4).

One way to cope with this problem is to compute the *realized half-spread*, that is, the difference between the transaction price and the midprice at some time, Δ , after the transaction (say, a few seconds later), where the interval Δ should be long enough to ensure that market quotes have adjusted to reflect the price impact of the transaction. Let p_t be the price of the transaction at time t , d_t the direction of the market order triggering it, and m_t the midprice at time t . The realized half-spread for this transaction is then given by:

$$S_r = d_t(p_t - m_{t+\Delta}) = d_t(p_t - m_t) - d_t(m_{t+\Delta} - m_t), \quad (2.5)$$

where the first term, $d_t(p_t - m_t)$, is the effective spread, and the second one, $d_t(m_{t+\Delta} - m_t)$, is the price impact of the transaction, defined as the (absolute) change in the midprice that occurs after it. The overall expression can thus be seen as a measure of the profit earned by the liquidity supplier on the transaction at time t if he unwinds his position at the midprice at $t + \Delta$. Using the definition of the effective spread in equation (2.3) in expression (2.5), one can rewrite the average realized bid-ask spread as:

$$E(S_r) = E(S_e) - E(d_t(m_{t+\Delta} - m_t)). \quad (2.6)$$

This expression shows that the average realized spread is smaller than the average effective spread if $E(d_t(m_{t+\Delta} - m_t)) > 0$, that is, if on average transactions have a positive price impact. Interestingly, if the effective spread is on average smaller than the price impact, liquidity providers would lose money on average, as $E(S_e) < E(d_t(m_{t+\Delta} - m_t))$. Since in the long run this would drive them out of the market, the effective spread cannot be too low: it must at least compensate them for the adverse price movement following a trade.

The value of the realized spread is sensitive to the choice of the reference post-trade market price (i.e., to $m_{t+\Delta}$). In practice, market participants need time to respond to the information content of transactions with fresh quotes and limit orders. Thus, the choice of Δ depends on how quickly market participants adjust their quotes after a transaction. In transparent and active markets (such as most equity markets), adjustment is generally fast, so a very small value of Δ is appropriate (Conrad and Wahal 2020): too high a value introduces unnecessary noise. In more opaque markets, such as the corporate bond market, information on trades takes more time to be disseminated, and higher values of Δ should be chosen.

Quoted, effective, and realized spreads are used routinely by market participants to evaluate market liquidity. Table 2.2 shows the average quoted, realized, and effective spread and price impact (in basis points) on shares of four large American companies over two trading days in 2020. Three points emerge from

Table 2.2 Average spread measures (in bps) from two trading days in 2020

Stock	March 16, 2020 (High volatility)			
	Quoted spread	Effective spread	Realized spread	Price impact
Apple	3.2	1.8	0.6	1.2
J.P. Morgan Chase	5.1	2.8	0.9	1.8
Pfizer	3.7	2.4	0.6	1.8
Tesla	18.0	6.5	2.3	4.1
Stock	August 5, 2020 (Low volatility)			
	Quoted spread	Effective spread	Realized spread	Price impact
Apple	1.0	0.5	0.1	0.4
J.P. Morgan Chase	1.3	0.9	0.3	0.6
Pfizer	2.6	1.2	0.6	0.5
Tesla	7.2	2.0	1.1	0.9

Note: The realized spread and price impact measures are computed over a time interval of 5 seconds.

Source: <https://research.wpcarey.asu.edu/investment-engineering/>.

the table. First, effective spreads are systematically smaller than quoted spreads, which reflects the fact that market participants place orders strategically at times when the quoted spread is particularly small: quotes are averaged over clock time, while effective spreads are averaged over transactions. Second, as expected, realized spreads are smaller than the corresponding effective spreads, which shows that on average market orders had a positive price impact. Thirdly, the spreads varied greatly between the two days. On March 16, 2020, when the broad U.S. stock index S&P 500 dropped by 12%, spreads were much wider compared to a less volatile trading day, such as August 5, 2020.

To see what the numbers in the table imply in terms of wealth transfers between market participants, consider a mutual fund buying \$1,000,000 worth of the Tesla stock on March 16, 2020, with a market order: given the effective spread, the fund would pay \$650 on the transaction, but only \$230 would be earned by the sellers of the stock if they turn around their position after 5 seconds, as following the transaction the Tesla stock will appreciate by 0.041%, so that \$410 would be lost due to price impact.

2.3 Other Measures of Implicit Trading Costs

The measures of trading costs considered so far require knowledge of bid and ask quotes at the time of execution (for the quoted spread), immediately before (for the effective spread) or shortly after (for the realized spread). Even when

quote data are unavailable, however—as is often the case—implicit trading costs can be measured using transaction data only, as shown below.

2.3.1 Volume-Weighted Average Price

When quote data are not available, the midquote cannot be used as the benchmark for the execution price. One can then use easily observable alternative benchmarks, such as the day's opening or closing price. But often averages of transaction prices over the day are used: a popular benchmark in trading cost analysis is the volume-weighted average price (VWAP) for all transactions in the stock over some relevant interval, normally the entire trading day (though one may also take a shorter interval around the time at which an order was placed). Whatever the benchmark time interval T chosen, we have:

$$VWAP = \frac{\$ \text{ volume of trading}}{\# \text{ of share traded}} = \sum_{t \in T} w_t p_t, \text{ where } w_t = \frac{|q_t|}{\sum_{t \in T} |q_t|}, \quad (2.7)$$

and q_t and p_t are the size and price of the t^{th} trade.

Investors evaluate their broker's performance in getting a good price for their order by comparing their own price with the day's VWAP. The difference is a mixture of the effective and the realized spreads, as it involves comparison with both pre- and post-transaction prices.

In practice, several problems arise with the use of VWAP as a benchmark for trading costs. First, the average may depend on the order one is interested in itself, if it accounts for a significant proportion of trading volume. In the extreme case, in which execution of a single order (in multiple fills) generates all the day's trades, then a measure of trading cost based on VWAP is automatically zero.

Second, VWAP-based measures of trading costs can be gamed by brokers. Consider, for instance, a broker in charge of executing a large buy market order. Typically, he splits the order and executes it in multiple fills, to avoid having too large an impact on prices. If his performance is judged by the VWAP, the broker has an incentive to trickle the order into the market extremely slowly to ensure that his average execution price is as close as possible to the VWAP. This could lead him to delay trade execution excessively, unduly increasing the risk of non-execution and the client's opportunity cost of not trading. As we will see in Section 2.4, clients can assess the opportunity cost of delay by applying the broader notion of implementation shortfall.

2.3.2 Measures Based on Price Impact

Other measures of transaction costs are based on the extent to which an order generates a reaction in the market price. As we saw in Section 2.2.3, the midprice tends to rise when buy orders arrive, to an extent that is positively correlated with their size. Symmetrically, it tends to fall in the wake of sell orders. Hence the change in the midprice in the wake of a transaction, Δm_t , can be seen as a measure of the price impact generated by that transaction. If this midprice change is proportional to the buying or selling pressure, the relationship can be expressed as follows:

$$\Delta m_t = \lambda q_t + \varepsilon_t, \quad (2.8)$$

where Δm_t is the change in the midprice over a fixed time interval (a half-hour, say, or a day) and q_t is the order (or “order flow”), that is, the total value of buy less sell market orders executed in the same interval. For instance, suppose that in a one-hour interval, the dollar value of all executed buy market orders is \$10,000, and that of sell orders \$8,000. The order imbalance is +2,000. This differs from trading volume, which in this case is eighteen thousand. Computation of the order imbalance again requires signing market orders. If data on the direction of buy and sell market orders is not available, one can use the Lee-Ready algorithm (Box 2.1).

The intuitive meaning of equation (2.8) is that the net demand during the chosen interval from traders placing market orders puts pressure on the price that is gauged by the coefficient λ . Chapter 4 provides the theoretical underpinning for this equation. The reciprocal of λ can be seen as a measure of market depth (see Chapter 1), in that a lower value of λ means prices are less sensitive to order imbalance. In practice, people often estimate this price impact measure λ by running a regression of the change in the midquote (Δm_t) on the order imbalance (q_t).⁵ A closely related approach is the price impact regression presented in Chapter 5.

Stoll (2000) applies a modified version of this approach to daily observations for stocks listed on the NYSE/AMEX and Nasdaq, including a lagged term (q_{t-1}) as an additional explanatory variable to capture a possible reversal of the price impact the following day. He finds that the value of λ is positive for 98% of the stocks and significantly different from zero for 63%. The measure varies considerably across stocks: a one-percentage-point increase in order imbalance has a price impact of 0.75% for lowest-capitalization NYSE/AMEX

⁵ Another approach is to regress the percentage change in the midquote on the order imbalance scaled by trading volume (i.e., the sum of buy and sell orders).

stocks and 0.52% for the highest-capitalization ones. So, as one would expect, larger stocks seem to have deeper markets, where the price better withstands order imbalances. For U.S. public debt markets, Adrian, Fleming, and Vogt (2017) estimate a price impact coefficient of 0.25 basis points for a \$100 million order imbalance in 2-year treasury notes, and of 2.02 basis points for 10-year notes over the 1991–2017 period.

Measuring order imbalances is sometimes difficult, as it requires data on the signed flow of buy and sell market orders. An alternative is to gauge the sensitivity of returns to trading volume (see Hasbrouck 2007, p. 93). While trading volume and order imbalance are certainly distinct concepts, they are likely to be correlated (days with larger order imbalances may well be the days with high trading volume). Therefore, one can estimate a regression of $|\Delta m_t|$ (the absolute value of price changes) on the trading volume Vol_t (the monetary value of the total amount traded) over the same interval (e.g., a day or a month). In this modified version, the slope can be interpreted as a measure of the price change associated with one additional unit of trading volume.

Conceptually this measure is related to the “illiquidity ratio” proposed by Amihud (2002), that is the ratio I_t of the absolute return for a stock ($|r_t|$) to trading volume over a given period (e.g., a day):

$$I_t = \frac{|r_t|}{Vol_t}. \quad (2.9)$$

This is also known as the Amihud ratio.⁶ The inverse of this measure,

$$L_t = \frac{Vol_t}{|r_t|}, \quad (2.10)$$

also frequently used, is known as the Amivest liquidity ratio. A low value of this ratio is a sign of market illiquidity.

2.3.3 Non-Trading Measures

Often the level of trading volume in a security or its turnover rate (volume divided by the market capitalization of the security) is used as an indicator of its liquidity—the idea being that a market where many buyers and many sellers congregate will offer better trading opportunities. But there is reason

⁶ Goyenko et al. (2009) show empirically that the Amihud ratio is a good proxy for price impact, as it is highly correlated with high-frequency measures of price impact.

to doubt that these are good measures of liquidity, because volume tends to increase when new information reaches the market, which is also a time of high volatility and concomitantly wide bid-ask spreads (as we shall see in Chapter 3). So, turnover may increase at a time when trading costs are high. For instance, Fleming (2003) finds that in the market for U.S. Treasury notes trading volume is a weak proxy for liquidity as measured by bid-ask spreads. He also finds that a trading frequency measure (namely, the number of trades executed in a specified interval) is a similarly poor proxy.

However, the frequency with which no trading at all occurs can be a useful proxy for illiquidity in some very thin markets, including many emerging markets. In such markets there are days or even weeks without transactions, in which the exchange simply reports the “stale price” of the last actual trade. Moreover, in these markets real-time quote data are generally unavailable. In a limit order market, this can happen because of the lack of counterparties; in a dealer market, because dealers’ bid-ask spreads are perceived as prohibitively costly or unrepresentative of actual trading opportunities.

As no trading is associated with no change in price, Lesmond, Ogden, and Trzcinka (1999) and Bekaert, Harvey, and Lundblad (2007) propose to measure illiquidity as the fraction of no-trade days, defined as days with zero daily returns. The advantage of this measure is that it requires only a time series of daily returns, and no transaction volume data. Bekaert, Harvey, and Lundblad (2007) and Lesmond (2005) reinforce the measure’s credibility, showing that it is positively correlated with bid-ask spreads for the limited periods when overlapping data are available, and negatively correlated with trading volume. They confirm this finding by comparing this illiquidity measure with more standard liquidity measures using U.S. data.⁷

Nevertheless, this measure also has a drawback: the maintained assumption that there is no trading whenever prices do not move is not always valid. For instance, on any given day the price of a security may stay level because there is no news about that security, even though trading does take place. Indeed it is precisely in a very liquid market that trades can take place without moving the price! To the extent that such cases occur, identifying non-trading with zero returns leads to an overestimate of the illiquidity of the corresponding market.

⁷ In their study, Bekaert, Harvey, and Lundblad (2007) also propose and construct another measure of illiquidity based on no-trade days. Their measure gives greater weight to no-trade days on which there are larger changes in the index of the market on which the stock is listed. The idea is that the change in the index should lead to a change in the price of individual stocks. When no such change is observed for a given stock, this is more likely to be due to absence of trading in this stock, possibly because of prohibitive trading costs.

2.3.4 Measures Based on Return Covariance

Roll (1984) sets out an ingenious method for measuring the bid-ask spread based on transaction prices alone. The idea is that orders will sometimes, at random, hit the ask and the bid price, so that transaction prices bounce back and forth between them, straddling the midquote m_t . The transitory deviations around the midprice are called bid-ask bounce. Intuitively, this bounce engenders negative serial correlation in transaction-to-transaction returns.

This is easiest seen by considering a security whose quotes stay unchanged over the relevant period. Suppose that a buy market order arrives at time t , followed by a sell market order at date $t + 1$. The first trade is at the ask price, the second at the bid price, so that the return between t and $t + 1$ is negative. Then, one can expect a positive return between $t + 1$ and $t + 2$, the time of the subsequent transaction: either the next market order is again a sell order executed at the bid price (in which case the return from $t + 1$ to $t + 2$ is zero) or it is a buy market order executed at the ask price (in which case the return from $t + 1$ to $t + 2$ is positive). A similar argument shows that after a positive return, one would expect a negative return.

Roll (1984) exploits this intuition to construct an estimator of the bid-ask spread, S , based entirely on the serial covariance of returns. This estimator is called Roll's measure and is derived below. The estimator depends on specific assumptions regarding the order arrival process. After presenting Roll's measure, we shall discuss how it should be adjusted when these assumptions do not hold.

Roll's Measure

To derive Roll's measure, suppose that the security's fundamental value, as captured by the midquote, follows a random walk:

$$m_t = m_{t-1} + \varepsilon_t, \quad (2.11)$$

where ε_t is mean-zero white noise ($E(\varepsilon_t) = 0$ for all t , and $E(\varepsilon_t \varepsilon_s) = 0$ for all $t \neq s$). This variable represents the change in the value of the stock due to new information emerging between time $t - 1$ and t . As $E(\varepsilon_t) = 0$, the expected fundamental return is zero ($E(m_t - m_{t-1}) = 0$), which is reasonable for small time intervals (a day or less). In any case this assumption can easily be relaxed (see below).

Now suppose that the bid-ask spread S is constant over time. The ask and bid prices in the market are:

$$a_t = m_t + \frac{S}{2} \text{ and } b_t = m_t - \frac{S}{2} \quad (2.12)$$

$$b_t = m_t - \frac{S}{2}. \quad (2.13)$$

Transactions are at either the bid or the ask price depending on whether a buy or a sell market order arrives. Thus, the price of the t^{th} transaction can be written:

$$p_t = m_t + \frac{S}{2}d_t, \quad (2.14)$$

where d_t ($= 1$ or -1) indicates whether a transaction is buyer or seller initiated (see Section 2.2.1). Therefore, using equations (2.11) and (2.14), the dollar transaction-to-transaction return is:

$$p_t - p_{t-1} = m_t + \frac{S}{2}d_t - \left(m_{t-1} + \frac{S}{2}d_{t-1}\right) = \frac{S}{2}d_t - \frac{S}{2}d_{t-1} + \varepsilon_t. \quad (2.15)$$

To compute Roll's estimator for the bid-ask spread, we need several additional assumptions on the order arrival process, namely:

- a. **Balanced order flow.** Market orders are equally likely to be buy or sell orders: $\Pr(d_t = 1) = \Pr(d_t = -1) = 1/2$ (so that $E(d_t) = 0$ for all t).
- b. **No autocorrelation in orders.** Buy and sell market orders are serially uncorrelated, i.e., $E(d_t d_s) = 0$ for $t \neq s$.
- c. **No effect on the midquote.** Market orders are assumed to carry no news, meaning that they are uncorrelated with current and future innovations in fundamentals: $E(d_t \varepsilon_t) = E(d_t \varepsilon_{t+1}) = 0$ for all t .
- d. **Constant (zero) expected return.** The fundamental value follows a random walk, so that $E(m_t - m_{t-1}) = E(\varepsilon_t)$ is constant and equal to zero for all t .

This set of assumptions regarding the order arrival process and the dynamics of price and quotes constitutes Roll's model. As shown in subsequent chapters, this model is an important element in many market microstructure models.

Under the assumptions of Roll's model:

$$E(p_t - p_{t-1}) = 0.$$

Therefore, using equation (2.15):

$$\begin{aligned} \text{cov}(p_{t+1} - p_t, p_t - p_{t-1}) &= E\left[\left(\frac{S}{2}d_{t+1} - \frac{S}{2}d_t + \varepsilon_{t+1}\right)\left(\frac{S}{2}d_t - \frac{S}{2}d_{t-1} + \varepsilon_t\right)\right] \\ &= \frac{S^2}{4}E[d_{t+1}d_t - d_t^2 - d_{t+1}d_{t-1} + d_t d_{t-1}] \end{aligned} \quad (2.16)$$

$$= -\frac{S^2}{4} \quad (2.17)$$

where we have used the assumptions on the order arrival process and the fact that $E(d_t^2) = 1$: as d_t can be only +1 or -1, its square is always 1. As expected, equation (2.17) implies that the bid-ask spread induces a negative correlation in price changes ($\text{cov}(p_{t+1} - p_t, p_t - p_{t-1}) < 0$ if $S > 0$). Moreover, it yields Roll's estimate of the absolute value of the bid-ask spread:

$$S_R = 2\sqrt{-\text{cov}(\Delta p_{t+1}, \Delta p_t)}, \quad (2.18)$$

also known as Roll's measure.

Stoll (2000) reports estimates of the (half) Roll's measure for all stocks listed on the NYSE and Nasdaq (see Stoll 2000, p. 1493, table III). For each stock, he calculates the average daily serial covariance in trade-to-trade price changes over sixty-one trading days. He then reports an estimate of $\sqrt{-\text{cov}(\Delta p_{t+1}, \Delta p_t)}$ by decile of capitalization. Overall, for the NYSE, the half-Roll's measure is 3.81 cents and 11.5 cents for Nasdaq. Obviously this difference is due to differences in the characteristics of the stocks listed on the two exchanges, but even after controlling for market capitalization, for instance, it persists. Stoll (2000), p. 1491, table II also reports the average values of the quoted and the effective spreads for his sample of stocks. On average, the half-quoted spread on the NYSE is 7.9 cents against 12.6 cents on Nasdaq; the half-effective spread, 5.6 against 10.7 cents. The effective spread is smaller than the quoted spread in both markets, since dealers in each sometimes offer price improvement to their clients, which results in trades inside the bid-ask spread. Roll's measure underestimates the quoted spread, and for the NYSE it also underestimates the effective spread.

Extensions

As noted, Roll's measure depends on a series of assumptions concerning the process of order arrival and on stock returns, without which it will not yield an unbiased estimate of the bid-ask spread. We now examine why and in which direction the failure of these assumptions biases Roll's measure. Let us relax them one at a time.

Assumption (a): balanced order flow. In reality, market buy and sell orders are not necessarily of equal probability. For instance, at the end of the trading day agents may be more anxious to close out short positions, in which case buyer-initiated transactions would be more prevalent. To account for this, suppose that $\Pr(d_t = 1) = \eta$. When η is different from $\frac{1}{2}$, the order flow is unbalanced: there are more buy orders than sell orders if $\eta > \frac{1}{2}$, conversely if $\eta < \frac{1}{2}$. In the

appendix to this chapter we show that in this case the autocovariance of price changes is:

$$\text{cov}(\Delta p_{t+1}, \Delta p_t) = \frac{S^2}{4} E[(d_{t+1} - d_t)(d_t - d_{t-1})] = -\eta(1 - \eta)S^2. \quad (2.19)$$

Therefore, an unbiased estimator of the bid-ask spread is:

$$S_a = \sqrt{-\frac{\text{cov}(\Delta p_{t+1}, \Delta p_t)}{\eta(1 - \eta)}}, \quad (2.20)$$

which implies:

$$S_R = (2\sqrt{\eta(1 - \eta)}) S_a.$$

Thus, unless the order flow is perfectly balanced ($\eta = \frac{1}{2}$), Roll's spread estimator is biased: It underestimates the true spread by a factor of $2\sqrt{\eta(1 - \eta)}$. To see intuitively why, note that if almost all transactions are buyer initiated (say, $\eta = 0.99$), the bid-ask bounce is almost nil, as almost all trades are at the ask price. Thus, the covariance of the change in prices is close to zero and Roll's estimate is also close to zero, even though the actual bid-ask spread may be large. Thus, one must adjust the estimate of the covariance (here multiplying it by a factor $[2\eta(1 - \eta)]^{-1} \approx 50.5$ to obtain a correct spread estimate.

Assumption (b): non-autocorrelated orders. Roll's method is also biased if there is serial correlation in the trade direction d_t . Such correlation may occur in practice because traders slice their orders and execute them piecemeal over time (see Chapters 3 and 9). For example, suppose the direction of market orders is first-order autocorrelated:

$$\Pr(d_{t+1} = d_t) = \delta.$$

We can then show (see appendix) that:

$$\text{cov}(\Delta p_{t+1}, \Delta p_t) = \frac{S^2}{4} E[(d_{t+1} - d_t)(d_t - d_{t-1})] = -(1 - \delta)^2 S^2. \quad (2.21)$$

Hence, an unbiased estimate of the bid-ask spread is given by:

$$S_b = \frac{1}{1 - \delta} \sqrt{-\text{cov}(\Delta p_{t+1}, \Delta p_t)} = \frac{S_R}{2(1 - \delta)}. \quad (2.22)$$

Thus, unless $\delta = \frac{1}{2}$ (the order flow is serially uncorrelated), Roll's estimator of the bid-ask spread underestimates the bid-ask spread by a factor of $2(1-\delta)$. Choi, Salandro, and Shastri (1988), using the Lee-Ready algorithm to classify transactions, estimate δ to be about 0.7, which implies positive serial correlation in the direction of trade. Then Roll's original measure would again underestimate the true spread by a factor of $2 \times 0.3 = 0.6$.

Assumption (c): no effect of orders on the midquote. Roll assumes that changes in the value of the security (ε_t) and the direction of market orders (d_t) are independent. This assumption is problematic if some traders submitting market orders have information on the stock's future payoff. In this case, as will be seen in Chapter 3, the direction of market orders does carry information, and market participants' estimate of the value the security moves in the direction of the order. Specifically, liquidity suppliers permanently revise their value estimate upward after buy orders and downward after sell orders, which implies that ε_t and d_t are positively correlated. This attenuates the bid-ask bounce, which means that Roll's estimator again underestimates the true spread. In this case, obtaining a more accurate estimator requires deeper understanding of the trading process in the presence of asymmetric information. The method will be presented in Section 5.2.

Assumption (d): non-varying fundamental expected return. One of the assumptions used to derive Roll's measure is that the expected return is constant and indeed equal to zero: $E(m_t - m_{t-1}) = 0$. However, this assumption may fail: expected returns for a security can vary over time, and if the variations are serially correlated this may be yet another source of bias in Roll's measure. To see this, suppose that:

$$m_t = m_{t-1} + \bar{r}_t + \varepsilon_t. \quad (2.23)$$

where the term $\bar{r}_t$ is the expected component of the return from time t to $t+1$ (that is, $\bar{r}_t \equiv E(m_t - m_{t-1})$) and ε_t is the unexpected component. We index $\bar{r}_t$ by t because it can be time-varying. Then, using equations (2.14) and (2.23), we obtain:

$$\Delta p_t = \bar{r}_t + \frac{S}{2}(d_t - d_{t-1}) + \varepsilon_t. \quad (2.24)$$

Hence,

$$\text{cov}(\Delta p_{t+1}, \Delta p_t) = \text{cov}(\bar{r}_{t+1}, \bar{r}_t) - \frac{S^2}{4}. \quad (2.25)$$

Thus, the covariance of change in prices now depend on two factors: (1) the bid-ask bounce $\left(-\frac{s^2}{4}\right)$ whose effect is to make change in prices negatively correlated, and (2) the covariance in expected returns $(\text{cov}(\bar{r}_{t+1}, \bar{r}_t))$. If the latter is non-zero, then Roll's estimate is biased again. By equation (2.25), an unbiased estimator of the bid-ask spread is:

$$S_d = 2\sqrt{-\text{cov}(\Delta p_{t+1}, \Delta p_t) + \text{cov}(\bar{r}_{t+1}, \bar{r}_t)}. \quad (2.26)$$

If expected returns are positively autocorrelated $(\text{cov}(\bar{r}_{t+1}, \bar{r}_t) > 0)$, S_d is larger than the Roll's estimator, which therefore again underestimates the true spread. George, Kaul, and Nimalendran (1991) discuss various ways of correcting for this problem through suitable time-varying estimators of $\bar{r}_t$.

The serial correlation in expected returns may explain why in some samples we find a positive covariance of price changes: in equation (2.25) this happens if $\text{cov}(\bar{r}_{t+1}, \bar{r}_t)$ is positive and large enough to swamp the bid-ask bounce effect. This was reported even in Roll's original 1984 study, which implements his estimator using daily returns (based on closing prices) and finds that roughly half of the cases show positive serial correlation.⁸ Then according to equation (2.18), the spread estimate would be the square root of a negative number—a nonsensical outcome! Harris (1990) shows that positive autocovariances are more likely for low values of the spread. Therefore, in such cases a rough remedy would be to impute a value of zero to the estimated spread. In general, Roll's estimator works better the shorter the time interval over which the returns are measured, so that bid-ask bounce is larger than other determinants of returns.

To sum up, the bid-ask spread induces negative serial correlation in returns, that is, it is a source of price reversals. The idea that illiquidity is a source of reversals is quite general, as Chapter 3 demonstrates. So, estimators for the bid-ask spread can be obtained by measuring the covariance of subsequent returns. These returns can be measured at high frequency (e.g., from trade to trade) or at lower frequency (e.g., from daily close to close).⁹ But the accuracy of these estimators depends on the properties of the order arrival and price processes, and they must be fine-tuned accordingly. For instance, exercise 4 asks the reader to derive an estimator of the bid-ask spread in Roll's model when some trades occur at the midprice, as sometimes happens in reality.¹⁰

⁸ In contrast, Stoll (2000) finds that for more than 99% of the stocks listed on Nasdaq and the NYSE, the covariance of trade-to-trade returns is negative, as Roll's model implies.

⁹ Nowhere do we impose that date t and date $t + 1$ correspond to successive transactions.

¹⁰ Another bias in Roll's estimator arises when there is no trade on a given day, in which case some data bases (such as CRSP) report the midquote instead of a transaction price. If these days are retained in the sample, the estimated cost will generally be biased downwards, because the midquote has no bid-ask bounce.

2.4 Implementation Shortfall

All the measures of trading costs presented so far are static, in that they do not take the time dimension of execution quality into account. In practice, however, large orders from institutions are very often split among several brokers over time (to maintain secrecy about the real size of the desired trade), and the brokers themselves may trickle the orders slowly into the market to get better prices. Thus, there can be a delay between the moment in which a portfolio manager makes his investment decision and the time he starts implementing it. This delay is costly if it is systematically associated with an adverse price movement (see the second column of Table 2.1 for an estimate of delay costs). Moreover, the portfolio manager also faces the risk of a large adverse price movement before the order is entirely filled, perhaps obliging him to cancel part of his order, which results in an opportunity cost.

For these reasons, portfolio managers often use a more encompassing measure of execution quality, namely the “implementation shortfall,” first proposed by Perold (1988). This gauge factors in not only the price impact of trades but also the opportunity cost of delayed or unexecuted orders. The basic idea is to benchmark the actual performance of a portfolio against a hypothetical “paper portfolio” in which all rebalancing trades are made instantaneously and fully at midprices. In these conditions, the difference between actual performance and that of the portfolio’s paper benchmark is precisely due to all the trading frictions (delays, price impact, and partial execution) encountered by the portfolio manager in rebalancing his portfolio. Thus, the concept of implementation shortfall enables us to disentangle, in the overall return of a portfolio, the portion deriving from the investment strategy of the manager and that depending on the implementation of these ideas, which is the responsibility of the trading desks hired for this purpose.

To understand how the measure of implementation shortfall is constructed and used, suppose that a portfolio manager decides to buy q shares of a stock on day 0 and passes the order to a broker. He plans to evaluate his position at some point in the future, say at date t (his “investment horizon”). In the absence of trading costs (i.e., in a perfectly liquid market), the portfolio manager would buy the shares at price m_0 and evaluate them at time t at price m_t . On paper, his dollar return is therefore:

$$R_p = q(m_t - m_0). \quad (2.27)$$

In reality, however, the portfolio manager will typically buy the stock at a higher price than m_0 because the market is illiquid. And as we have seen, part of the order may go unexecuted. For instance, the manager may decide to stop buying

if the price goes too high. If $m_t > m_0$, this results in an opportunity cost. To account for these costs, suppose that only a fraction κ of the manager's buy order is eventually filled, at average execution price $\bar{p}$. Thus, the actual return is:

$$R_a = \kappa q(m_t - \bar{p}). \quad (2.28)$$

The implementation shortfall, IS , is the difference between the portfolio's theoretical, paper return and the actual return:¹¹

$$IS \equiv q(m_t - m_0) - \kappa q(m_t - \bar{p}) = \underbrace{\kappa q(\bar{p} - m_0)}_{\text{execution cost}} + \underbrace{(1 - \kappa)q(m_t - m_0)}_{\text{opportunity cost}}. \quad (2.29)$$

Expression (2.29) shows that the implementation shortfall consists of two components: execution cost and opportunity cost. The first term should be familiar by now: it captures the cost of the actual execution at the average price $\bar{p}$ rather than the benchmark price m_0 (i.e., the effective spread). The second component represents the opportunity cost of the forgone returns on the unexecuted portion of the order, $(1 - \kappa)q$. For example, imagine that a client submits a buy order for ten thousand shares when the midquote is \$100, and evaluates the performance of his broker two days later, when the broker has only bought three thousand shares at an average price of \$101.¹² If the current price is 103, then the implementation shortfall is:

$$3,000 \times (101 - 100) + 7,000 \times (103 - 100) = 3,000 + 7,000 \times 3 = 24,000.$$

To put this figure in perspective, consider that the initial value of the “paper” portfolio in this example is $qm_0 = 1,000,000$, so that the implementation shortfall is 2.4% of its value.

As observed earlier, there may be a delay between the moment when the portfolio manager makes the investment decision and the moment of implementation. The execution cost component of the implementation shortfall can itself be divided in two components to reflect the cost of this delay. To see this let τ be the date at which the buy order begins to be executed. Then, the execution cost component can be written:

¹¹ The implementation shortfall is computed symmetrically for a sell order, but in that case q is negative (q being the signed order size).

¹² Frazzini, Israel, and Moskowitz (2018) use a sample of large orders submitted by a large asset manager. The average order size of this manager (q) varies between 0.1% to 13% of the average daily volume and it takes about 2.7 days to execute an order.

$$\kappa q(\bar{p} - m_0) = \kappa q(\bar{p} - m_\tau) + \kappa q(m_\tau - m_0), \quad (2.30)$$

where the second element $\kappa q(m_\tau - m_0)$ is the delay portion of the execution component of the implementation shortfall. Table 2.1 shows that in practice the delay portion accounts for a significant fraction of total execution cost.

For any given order, one or all components of the implementation shortfall may be negative, as when the broker manages to fill the order at a better average price than the initial price of the paper portfolio. This may happen when the price of the stock goes down after a buy order is issued, or alternatively if the broker is able to use limit rather than market orders. The opportunity cost is negative if the value of the stock declines over the investment horizon ($m_t < m_0$). In hindsight, failing to buy the security turns out to be the better choice and the opportunity cost is negative.

As with the other measures of trading costs, the average implementation shortfall over a large number of orders is a more meaningful gauge than its value for any given order, in that the average filters out the impact of random price variations. The average implementation shortfall is:

$$E(IS) \equiv \kappa E(q(\bar{p} - m_0)) + (1 - \kappa) E(q(m_t - m_0)).$$

The opportunity cost component will be positive on average if the direction and size of the orders placed (q) is positively correlated with price changes over the portfolio manager's investment horizon ($m_t - m_0$) so that $E(q(m_t - m_0)) > 0$. In practice, this positive correlation may exist for several reasons: one's manager may trade based on the same signal as other market participants, the order may exert pressure on prices, or some traders may "front-run" the order (i.e., buy the security to resell it to the buyer at a higher price). Further, brokers often face a trade-off between the execution and the opportunity cost components. By trading patiently (as with limit orders), a broker can generally get a better price and so reduce the execution cost component. But as this strategy increases the risk of non-execution ($1 - \kappa$), it results in a larger opportunity cost component on average.

The optimal strategy given this trade-off depends on the broker's forecast of the dynamics of the midprice and market liquidity. Again consider a broker with a large buy order. Obviously, if he expects the price to increase, he should increase the execution speed. But, as Chapter 1 shows, buy market orders deplete liquidity on the sell side. In general, after a while, liquidity suppliers will post new quotes and the market will become more liquid. Resiliency, the speed at which liquidity returns to normal after a trade (e.g., how soon a limit order is replenished after the arrival of a large market order), is another dimension of market liquidity. If the resiliency of the market is high, the broker can accelerate

the execution of his large order; if resiliency is low, he should trade more slowly. In highly resilient markets, therefore, brokers will be able to achieve both lower execution costs and lower opportunity costs.

In practice, the amount of the implementation shortfall is not negligible in relation to portfolio returns.¹³ For this reason, institutional investors increasingly focus on trading cost management in the effort to reduce this cost, often with the help of consulting services from brokerage houses, which now commonly provide so-called trading cost analysis (TCA) services to portfolio managers. Typically, TCA inquires into the sources of the implementation shortfall on a portfolio manager's orders.

2.5 Hands-On Estimation of Transaction Costs

We have presented many measures of transaction costs. The best way to appreciate similarities and differences is for the reader to estimate a few of them and see how their values depend on the characteristics of the stock and how they change in the course of the trading day. For this, we provide data from real-world securities markets, as training grounds for hands-on estimation. In the exercises of this chapter, the reader is prompted to use these data, some of which are available in the companion web site for the book.

2.6 Further Reading

Holden, Jacobsen, and Subrahmanyam (2013) offer an in-depth review of liquidity measures and empirical evidence on liquidity. The literature has proposed several other measures of illiquidity, in addition to those presented here. The “effective tick” proposed by Holden (2006) is based on the idea that price observations tend to cluster at certain ticks, specifically at rounder increments (price clustering), only due to the bid-ask spread. The measure of the spread is obtained in two steps: first, one estimates the probability of each bid-ask spread from the observed frequencies of prices at the various ticks; second, these probabilities are used to compute the probability-weighted average of each effective spread size.

Another proposed estimator of the effective spread, the LOT measure Lesmond, Ogden, and Trzcinka (1999), is based on the idea that if no trading

¹³ For instance, Leinweber (1995) reports that the paper return of the Value Line Portfolio (based on the recommendations of the Value Line newsletter) was 26.2% over the period 1971–91 while the actual return was 16.1%, implying an implementation shortfall of 10.1%. Frazzini, Israel, and Moskowitz (2018) estimate the implementation shortfall for actual trades of a large asset manager at 11.8 bps (per order) on average over the 1998–2016 period.

occurs on zero-return days, informed traders must have faced transaction costs exceeding the price change that their information implies. Therefore, the authors propose a maximum likelihood estimator for the transaction cost corresponding to the observed no-trading price interval.

Hasbrouck (2002) sets out two alternative methods for estimating Roll's measure. He notes that in Roll's model both the spread, S , and the variance of the change in the value of the security, $\text{var}(\varepsilon_t)$, are unknown parameters. They can be estimated either with the classical method-of-moments or with Bayesian techniques, which involve the specification of a prior. An argument for the Bayesian approach in this context is its ability to accommodate latent (unobserved) data, which in Roll's model include bids, asks, and trade direction indicators, and which are suppressed in the GMM estimation. Since Hasbrouck implements this Bayesian estimation via an iterative procedure known as the Gibbs sampler, the resulting estimate of the spread is known as "Gibbs estimate."

Hasbrouck (2009) compares the Gibbs estimate based on daily closing prices from CRSP with the effective cost of trading based on single-transaction trade and quote data from the TAQ database over the period 1993–2005, and finds that they are highly correlated. This suggests that less data-intensive estimates obtained at the daily frequency are a good approximation of the more precise estimates that can be obtained from intra-daily-frequency data.

Goyenko, Holden, and Trzcinka (2009) effect a similar comparison for a larger set of measures of effective spread and price impact, investigating how closely the measures computed at monthly and annual frequencies and based on daily data are correlated with the corresponding estimates based on intra-daily data from TAQ and rule 605 data. Again, their exercise is designed to identify the measures that, when computed on daily data, best approximate the behavior of their more data-intensive counterparts.

Abdi and Ranaldo (2017) propose another method to estimate the bid-ask spread that improves on Roll's measure by exploiting closing, high, and low prices (as opposed to closing prices only). The advantage of their method is that it is independent of trade direction dynamics, and thus does not require the assumption of serially independent and equally likely trade directions. Instead of the autocovariance of consecutive close-to-close price returns as in Roll (1984), their estimator relies on the autocovariance of close to mid-range returns around the same close price, where mid-range prices are defined as the mean of the daily high and low log-prices.

Determining what trading strategies are optimal to minimize implementation shortfall has become an important issue for practitioners: Bertsimas and Lo (1998) lay out the foundations for this type of analysis; a systematic treatment can be found in Kissell and Gantz (2003). Frazzini, Israel, and Moskowitz (2018) provide examples of execution strategies for orders of a large asset manager

(e.g., how these strategies optimally combine limit and market orders) and report estimates the implementation shortfall for on these orders.

2.7 Appendix. Extensions of Roll's Measure

We derive the expressions for the various estimators of the bid-ask spread in Section 2.3.4. Let $\Delta d_{t+1} \equiv d_{t+1} - d_t$. Recall that:

$$\begin{aligned} \text{cov}(\Delta p_{t+1}, \Delta p_t) &= \text{cov}\left[\left(\frac{S}{2}d_{t+1} - \frac{S}{2}d_t + \varepsilon_{t+1}\right), \left(\frac{S}{2}d_t - \frac{S}{2}d_{t-1} + \varepsilon_t\right)\right] \\ &= \text{cov}\left(\frac{S}{2}\Delta d_{t+1} + \varepsilon_{t+1}, \frac{S}{2}\Delta d_t + \varepsilon_t\right) \\ &= \left(\frac{S}{2}\right)^2 \text{cov}(\Delta d_{t+1}, \Delta d_t) + \left(\frac{S}{2}\right) \text{cov}(\Delta d_{t+1}, \varepsilon_t) \\ &\quad + \left(\frac{S}{2}\right) \text{cov}(\varepsilon_{t+1}, \Delta d_t) + \text{cov}(\varepsilon_{t+1}, \varepsilon_t). \end{aligned} \quad (2.31)$$

a) Relaxing the assumption of balanced order flow

Consider the case in which the order flow is unbalanced, that is, $\Pr(d_t = 1) = \eta$. Other assumptions are as in Roll's baseline model. In this case, using equation (2.31), we have:

$$\text{cov}(\Delta p_{t+1}, \Delta p_t) = \left(\frac{S}{2}\right)^2 \text{cov}(\Delta d_{t+1}, \Delta d_t).$$

By definition:

$$\text{cov}(\Delta d_{t+1}, \Delta d_t) = E(\Delta d_{t+1} \Delta d_t) - E(\Delta d_{t+1})E(\Delta d_t). \quad (2.32)$$

To calculate $\text{cov}(\Delta d_{t+1}, \Delta d_t)$, consider Table 2.3, which shows the various possible realizations for (d_{t-1}, d_t, d_{t+1}) and their probabilities. The table thus allows us to calculate $E(\Delta d_{t+1} \Delta d_t)$ and $E(\Delta d_{t+1})E(\Delta d_t)$. Hence:

$$\begin{aligned} E(\Delta d_{t+1} \Delta d_t) &= \eta^2(1 - \eta)(-2)(+2) + \eta(1 - \eta)^2(+2)(-2) \\ &= \eta(1 - \eta)(\eta + 1 - \eta)(-4) = -4\eta(1 - \eta), \end{aligned}$$

and:

$$E(\Delta d_t) = \eta(1 - \eta)(-2) + \eta(1 - \eta)(+2) = 0,$$

Table 2.3 Probability, Value, and Change of Trade Direction, if $\Pr(d_t = 1) = \eta$

$\Pr(d_{t-1})$ d_{t-1}	η +1				$(1 - \eta)$ -1			
$\Pr(d_t)$ d_t Δd_t	η +1 0	$(1 - \eta)$ -1 -2			η +1 +2	$(1 - \eta)$ -1 0		
$\Pr(d_{t+1})$ d_{t+1} Δd_{t+1}	η +1 0	$(1 - \eta)$ -1 -2	η +1 +2	$(1 - \eta)$ -1 0	η +1 0	$(1 - \eta)$ -1 -2	η +1 +2	$(1 - \eta)$ -1 0

so that:

$$\text{cov}(\Delta d_{t+1}, \Delta d_t) = E(\Delta d_{t+1} \Delta d_t) = -4\eta(1 - \eta).$$

Substituting this expression into (2.31) and using the assumptions that the order flow and change in asset values are uncorrelated, we obtain:

$$\text{cov}(\Delta p_{t+1}, \Delta p_t) = -4\eta(1 - \eta) \frac{S^2}{4} = -\eta(1 - \eta) S^2,$$

which is equation (2.19).

b) Relaxing the assumption of uncorrelated order flow

Now consider the case in which the direction of orders is first-order autocorrelated, with constant transition probability, $\Pr(d_{t+1} = d_t) = \delta$, while other assumptions are as in the baseline Roll model. To analyze this case, we have to modify the previous table as follows:

From Table 2.4, we obtain:

$$\begin{aligned} E(\Delta d_{t+1} \Delta d_t) &= \frac{1}{2}(1 - \delta)^2(-2)(+2) + \frac{1}{2}(1 - \delta)^2(+2)(-2) \\ &= (1 - \delta)^2(-4) = -4(1 - \delta)^2 \end{aligned}$$

and:

$$E(\Delta d_t) = \frac{1}{2}(1 - \delta)(-2) + \frac{1}{2}(1 - \delta)(+2) = 0.$$

Hence, using equation (2.32),

$$\text{cov}(\Delta d_{t+1}, \Delta d_t) = E(\Delta d_{t+1} \Delta d_t) = -4(1 - \delta)^2.$$

Table 2.4 Probability, Value, and Change of Trade Direction, if $\Pr(d_{t+1} = d_t) = \delta$

$\Pr(d_{t-1})$	$\frac{1}{2}$ +1				$\frac{1}{2}$ −1			
d_{t-1}								
$\Pr(d_t)$	δ		$(1-\delta)$		$(1-\delta)$		δ	
d_t	+1		−1		+1		−1	
Δd_t	0		−2		+2		0	
$\Pr(d_{t+1})$	δ	$(1-\delta)$	$(1-\delta)$		δ	$(1-\delta)$	$(1-\delta)$	δ
d_{t+1}	+1	−1	+1	−1	+1	−1	+1	−1
Δd_{t+1}	0	−2	+2	0	0	−2	+2	0

Substituting this expression into (2.31) yields:

$$\text{cov}(\Delta p_{t+1}, \Delta p_t) = -4(1-\delta)^2 \frac{S^2}{4} = -(1-\delta)^2 S^2,$$

which is equation (2.21).

2.8 Exercises

- Quoted spread for different trade sizes.** Consider the LOB of stock XYZ in Table 2.5, which displays the prices, sizes (i.e., number of shares) and times of entry of limit orders present in the LOB, as in Figure 1.2 of Chapter 1. From these data, compute the weighted average quoted spread (in absolute and relative terms) for 100, 500, 1,000, and 2,000 shares. Which side of the LOB is deeper for transactions of 2,000 shares or more?
- Measures of the bid-ask spread.** Your fund is considering trading 10-year bonds issued by the Austrian government, and you see that at 9:30 a.m. their lowest ask price is 102.31 and their highest bid price is 99.50. Five seconds later a buy order for a block of €10 million is executed at 102.76. At 10:30 a.m. you check the market again and see that the lowest ask price is 102.55 and the highest bid price is 100.02.
 - Compute the absolute and the relative quoted spread at 9:30 and 10:30.
 - Compute the absolute and the relative effective ask-side half-spread at 9:30.
 - Compare the quoted half-spread with the effective ask-side half-spread (both in absolute and in relative terms) at 9:30. What explains the difference between them?
 - Compute the absolute realized spread in the 9:30–10:30 interval.

Table 2.5 LOB OF STOCK XYZ

Bid			Ask		
Price	Size	Time	Price	Size	Time
74.42	300	11:49:39	74.48	300	11:49:35
74.41	100	11:46:55	74.48	500	11:49:40
74.36	400	11:48:30	75.74	100	08:25:17
74.36	400	11:48:32	76.00	150	08:02:02
74.00	13	10:56:00	76.77	20	07:01:01
73.75	5,100	11:28:02	77.00	100	09:15:00
72.98	5,100	10:57:39	77.06	200	10:14:11
72.15	120	08:01:39	77.35	1,000	08:01:39
72.00	20	07:01:01	77.82	20	07:01:01
72.03	20	07:01:01	78.00	300	08:02:00
72.00	100	07:46:19	78.38	1,000	09:30:04
71.59	50	08:02:02	78.60	375	08:01:32
71.11	20	07:01:01	78.64	500	09:30:04
71.00	10	09:30:36	78.87	20	07:01:01
70.35	200	08:00:54	78.95	200	08:01:35
70.11	20	07:01:01	80.00	350	09:15:00

- e. Compare the realized spread computed under point **d** with the absolute effective spread at 9:30 computed under point **b**. What explains the difference between them?
- 3. Implicit bid-ask spread in call auction.** In a call auction there is no bid-ask spread, as all trades clear at a single price. However, there is an implicit spread, insofar as the order flow exerts price pressure: the difference between the hypothetical prices that would clear the market if one tried to buy and sell more shares. Specifically, the implicit bid-ask spread for a transaction of size q can be defined as the difference in market clearing price arising from an extra market order of size q to buy and an extra order of size q to sell. Using the data in exercise 1 of Chapter 1, compute the bid-ask spread for transaction sizes $q = 50, 150, 250$, and 350 .
- 4. Roll's estimator and price improvements.** Consider Roll's model presented in Section 2.3.4. All the assumptions are unchanged, but we do assume that the transaction occurring at time t occurs either at the ask or bid price with probability λ , or at the midprice with probability $1 - \lambda$. Thus, $1 - \lambda$ can be seen as the fraction of trades that receive a price improvement or are crossed by brokers at the midprice. Propose an estimator of the quoted bid-ask spread S for this case.
- 5. Empirical measurement of quoted spreads.** The data for this exercise are contained in an Excel file, Ch2_AGF_data.xls available in the companion

website for the book: a record of one day's transactions in the shares of a French company, AGF, on the Paris Bourse. The data comprise:

- time of the transaction
 - size of the transaction
 - (average) transaction price
 - best bid price immediately before the transaction
 - best ask price immediately before the transaction
 - direction of trade initiation (-1 for transactions below the midprice, $+1$ above the midprice)
- a. For each transaction, compute the absolute spread S in euro (€); compute the relative spread, s and the log spread: $\ln(\text{ask}) - \ln(\text{bid})$; and compare the average of these three measures. Then compute the average absolute spread for each of the 17 half-hour time periods of the trading day and plot a graph of your results showing the intraday evolution of the spread. What kind of pattern over the day would you expect, a priori?
 - b. Compute and compare the average effective trading cost or “half-spread” in absolute terms, relative terms, and logs.
 - c. Compute the VWAP for the day. Then calculate the VWAP for buyer- and seller-initiated transactions for the whole day and compare your results with the VWAP benchmark. Repeat, again separately for buyer- and seller-initiated transactions, for transactions divided into three time periods: 9:00–12:00 a.m., 12:00–3:00 p.m., and 3:00–5:30 p.m.
 - d. Compute Roll's estimate of the bid-ask spread both in euro (€) and in relative terms (using the logarithm of the prices). Then repeat the computations in clock time rather than in transaction time: take the last transaction in every 15-minute time interval. Compare your results with those previously obtained: what explains the difference in Roll's measure?
 - e. Split the trading day into 15-minute intervals: 9:00–9:15 a.m., 9:15–9:30 a.m., . . . , 5:15–5:30 p.m. For each interval, compute the midprice change (from the last transaction of the previous interval to the last one of the current interval; for the first interval take the midprice at 9:06:04 as the initial midprice) and the cumulative signed order flow over the interval expressed as a fraction of the day's total (unsigned) order flow. Perform a regression analysis on the thirty-four data points you obtain in this way, to estimate the price impact parameter λ . Is the estimated parameter significantly different from zero? What is the impact on the midprice of a 1% relative order flow increase?
- 6. Inferring trade direction.** A data set containing a 1-day time series of quote and trade data for Krispy Kreme, listed on the NYSE, is provided

in the Excel file Ch2_KrispyKreme_raw_data.xls available on the companion website for the book. This data set contains both transaction data and quote revisions, but no trade direction indicator. (The variable “type” is equal to 1 for transaction data and 0 for quote data.)

- a. For each quote revision, compute the absolute spread S in dollars (\$), the relative spread s , and the log spread. Then compute the average absolute spread for each of the 14 half-hour time periods of the trading day and plot a graph of your results showing the intraday evolution of the spread.
 - b. Use the Lee-Ready trade classification algorithm to establish trade direction. Then generate a data set containing only transaction data (for each transaction consider the last bid price and the last ask price before the transaction takes place) of the same form as the AGF data set.
- 7. Further empirical transaction cost measurement.** Using the data set generated at point **b** of exercise 6 (or alternatively the data in the Excel file Ch2_KrispyKreme_data.xls, which also contains the trade direction indicator):
- a. Compute and compare the average effective trading cost or “half-spread” in absolute terms, relative terms, and logs.
 - b. Compute the VWAP for the day. Then calculate VWAPs for buyer- and seller-initiated transactions for the whole day and compare your results with the VWAP benchmark. Repeat, again separately for buyer- and seller-initiated transactions, for transactions divided into three time periods: 9:30–12:00 a.m., 12:00–2:00 p.m., and 2:00–4:00 p.m.
 - c. Compute Roll’s estimate of the bid-ask spread both in dollar (\$) and in relative terms (using the logarithm of the prices); then repeat the computations using not transaction time but clock time: take the last transaction in every 15-minute time interval.
 - d. Split the trading day into 15-minute intervals: 9:30–9:45 a.m., 10:00–10:15 a.m., . . . , 3:45–4:00 p.m. For each interval, compute the midprice change (from the last transaction of the previous interval to the last one of the current interval; for the first interval take the midprice at 9:30:02 as the initial midprice) and the cumulative signed order flow over the interval expressed as a fraction of the day’s total (unsigned) order flow. Perform a regression analysis on the 26 data points you obtain in this way, to estimate the price impact parameter λ . Is the estimated parameter significantly different from zero? What is the impact on the midprice of a 1% relative order flow change?
- 8. Implementation shortfall.** Suppose that at time 0 your brokerage firm receives an order from a client who would like to buy a number q of

shares in company XYZ, planning to hold them until some future time t . The midprice of XYZ is m_0 when you receive the order, and the client expects its midprice to be m_t at time t . Although in a perfectly liquid market the client would like the entire purchase to go through at time 0, he realizes that this may not be in his best interest in the rather illiquid market for XYZ shares. So the client's mandate to your firm is "choose the fraction k of the total purchase q that is to be bought between time 0 and time t so as minimize the implementation shortfall." You know that the average price $\bar{p}$ at which you can buy shares for this client between time 0 and t is affected by how much you buy in that interval, according to the function:

$$\bar{p} = m_0 + \lambda kq,$$

where λ is a price pressure parameter.

- a. Write the expression for the implementation shortfall in this specific case (using the above expression for $\bar{p}$).
 - b. Determine the value of k that minimizes the implementation shortfall.
 - c. How does the optimal k found under point **b** respond to changes in $m_t - m_0$, in q and in λ ? What are the intuitive explanations for these comparative statics?
- 9. Choice between limit and market orders.** Suppose you wish to buy 100 shares of a stock whose current price is €25 per share, and you are uncertain whether you should place a market order for 100 shares or a buy order with limit price at €26 for 100 shares. You turn for advice to your broker, telling him that you wish to minimize execution risk. He tells you: "it is not clear whether you are better off with either strategy." Is he right, in principle?
- 10. Order placement strategy and execution risk.** Consider a security with a final payoff v , which can be traded at time $t = 1$ and/or $t = 2$ at prices:

$$p_t = v + \lambda_t q_t + \epsilon_t, \quad (2.33)$$

where p_t is the price obtained by a customer who places an order of size q_t and λ_t is the price impact parameter, and an innovation ϵ_t that captures the arrival of "news" on the market at time t . The innovation ϵ_t has $E(\epsilon_t) = 0$ and $\text{var}(\epsilon_t) = \sigma_\epsilon^2 > 0$, and is serially uncorrelated.

Consider a risk-averse buyer with mean-variance utility function:

$$U(\Pi) = E(\Pi) - \gamma \text{var}(\Pi), \quad (2.34)$$

where Π is the investor's payoff. The investor assigns value v to each unit of the security, and wishes to place a total order of size Q , so that his payoff is:

$$\Pi = vQ - (q_1 p_1 + q_2 p_2). \quad (2.35)$$

At $t = 1$ he knows the current price of the security p_1 : so both $E(\Pi)$ and $\text{var}(\Pi)$ in (2.34) are conditional on the actual value of p_1 , while p_2 is uncertain for the investor.

The investor must decide how to split the total order Q between time 1 and time 2, namely determine q_1 and $q_2 = Q - q_1$ so as to maximize (2.34), where Π is given by (2.35) and prices p_1 and p_2 are given by (2.33).

- a. Compute the optimal order sizes q_1^* and q_2^* .
- b. Consider the “average split” of the total order Q over time, i.e.:

$$\frac{E(q_1^*)}{Q} \quad \text{and} \quad \frac{E(q_2^*)}{Q},$$

- recalling that, from an ex ante perspective, $E(\epsilon_1) = 0$. (i) What would this average split be if the investor were risk neutral ($\gamma = 0$)? Explain the intuition for this “risk-neutral average split.” Benchmark your analysis against the case $\lambda_1 = \lambda_2 = \lambda$, where the price pressure is the same in the two periods. (ii) Now consider a risk averse investor ($\gamma > 0$): how does risk aversion affect the average split of the total order Q ? Does it accelerate or delay order execution? Explain the intuition for this result, again using the case $\lambda_1 = \lambda_2 = \lambda$ as benchmark.
- c. Now consider the “actual split” of the total order Q over time, i.e.:

$$\frac{q_1^*}{Q} \quad \text{and} \quad \frac{q_2^*}{Q},$$

and suppose for concreteness that the price shock at $t = 1$ is positive ($\epsilon_1 > 0$), so that the security is more expensive than usual in the first period. Explain intuitively how this affects the actual split of the total order Q over time, and why.